

Task 1:

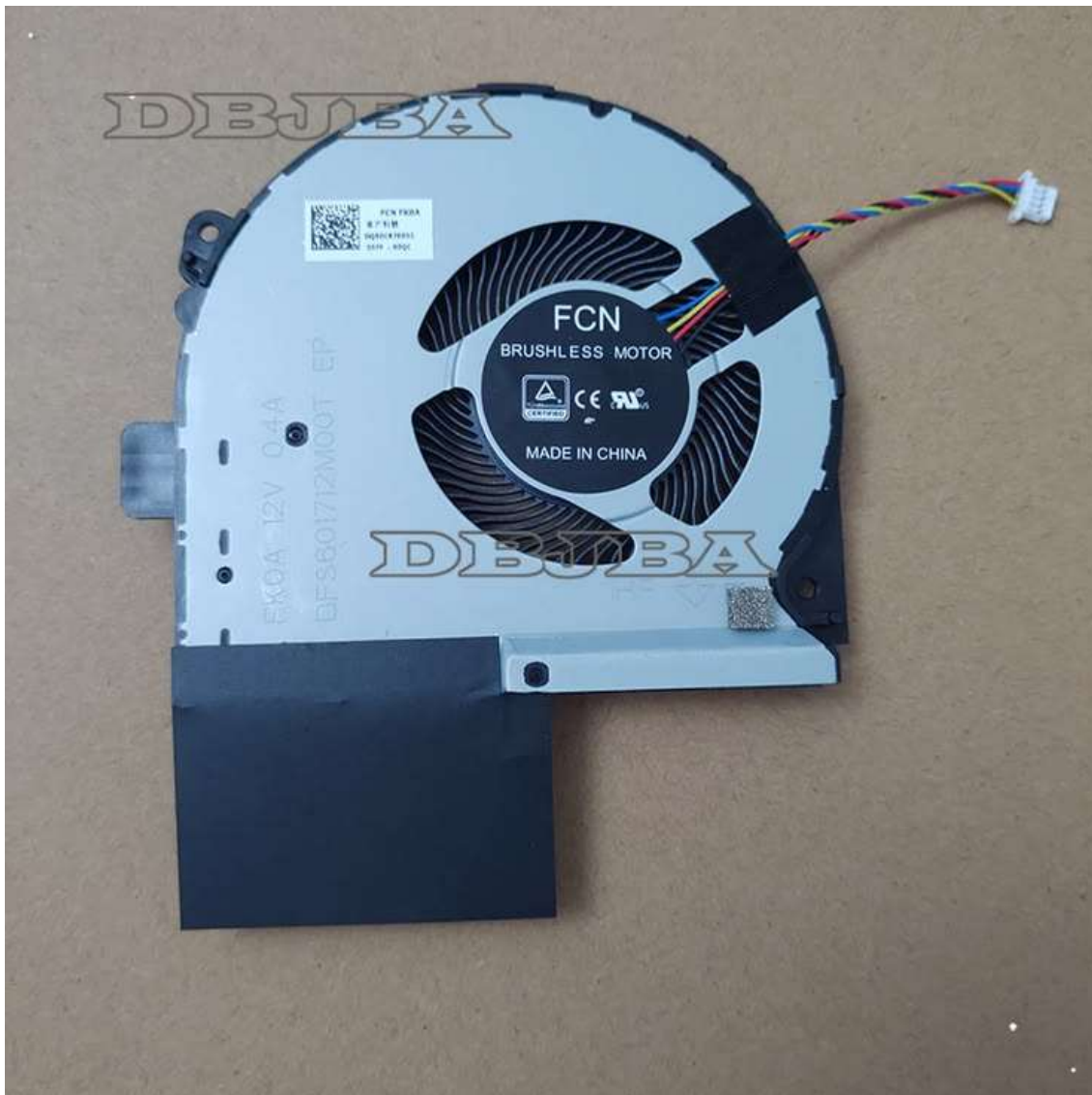
Identifying the GPU Model

have successfully checked and identified the graphics processing unit installed on my system using the Windows Task Manager utility. Below are the hardware specifications captured from my system:

- **GPU Brand:** Intel
- **Model Number:** Intel(R) UHD Graphics
- **Type:** Integrated GPU (iGPU)
- **System Detail:** The hardware configuration shows a Shared GPU Memory system, which dynamically utilizes the main system RAM to handle graphical processing tasks rather than relying on a separate dedicated video memory chip.

Task 2:

Cooling System Identification



1. CPU Heatsink:

The solid metal block (usually made of aluminum or copper fins) that sits directly over the processor. Its primary function is to absorb the massive heat generated by the CPU chip.

2. Cooling Fan (Active Cooler):

The rotating fan mounted securely on top of the heatsink. Its function is to push cool ambient air forcefully through the metal fins, blowing the heat away to prevent the system from thermal throttling and freezing.

Task 3:

Technical Comparison of Air Cooling and Liquid Cooling Systems

Feature / Criteria	Air Cooling System (Heatsink + Fan)	Liquid Cooling System (AIO / Custom Loop)
Working Mechanism	Uses conductive metal heat pipes (copper/aluminum) to pull heat into fins, which is then blown away by high-RPM active fans.	Uses a closed-loop system where a motorized pump circulates liquid coolant to absorb heat and transfer it to a radiator.
Cooling Efficiency	Highly efficient for standard	Outstanding efficiency.

Feature / Criteria	Air Cooling System (Heatsink + Fan)	Liquid Cooling System (AIO / Custom Loop)
	and mid-range workloads. Can experience thermal saturation under continuous heavy loads.	Dissipates heat much faster, keeping temperatures significantly lower even during extreme usage.
Acoustic Performance (Noise)	Can become quite loud when the fans have to spin at maximum speed under high computational strain.	Generally much quieter because the large surface area of the radiator requires lower fan speeds to cool the liquid.
Cost and Maintenance	Very cost-effective, easy to install, and	Premium pricing. Requires

Feature / Criteria	Air Cooling System (Heatsink + Fan)	Liquid Cooling System (AIO / Custom Loop)
	requires zero long-term maintenance other than occasional dust cleaning.	careful installation and involves risks of pump failure or liquid leakage over time.

Task 4:

High-Performance Scenario Analysis and Preventative Maintenance

When a gaming PC is broadcasting or streaming live content in 4K resolution, both the processor (CPU) and the graphics card (GPU) run at maximum workload capacities. The system processes massive amounts of high-bitrate video encoding data, which rapidly generates intense thermal energy. Without proper heat dissipation, the hardware will trigger thermal throttling, leading to sudden frame drops, buffering, or system crashes.

Mandatory Preventative Maintenance Protocol

Checklist: To keep the system running efficiently under high-performance scenarios, a technician should regularly execute the following checklist:

1. **Power Isolation & Safety:** Always completely shut down the computer, turn off the SMPS toggle switch,

and disconnect the main AC wall cable before working on the internal chassis.

2. Dust Cleanout: Use compressed air or an anti-static blower to thoroughly clean out dust accumulation locked between the fine gaps of the CPU heatsink fins, the GPU cooling shroud, and the chassis case filters to restore optimal airflow paths.

3. Thermal Paste Replacement: Carefully clean old, dried-up thermal compound from the processor chip using 99% Isopropyl Alcohol (IPA) and apply a fresh, high-grade pea-sized drop of thermal paste before securely remounting the cooler assembly to ensure perfect thermal conductivity.

4. Mechanical Fan Inspection: Manually rotate all system intake and exhaust fans to check for mechanical grinding noise, resistance, or bearing damage to ensure they can sustain high-RPM operations smoothly under maximum system load.